Bangladesh Army University of Engineering & Technology

Department of Computer Science and Engineering

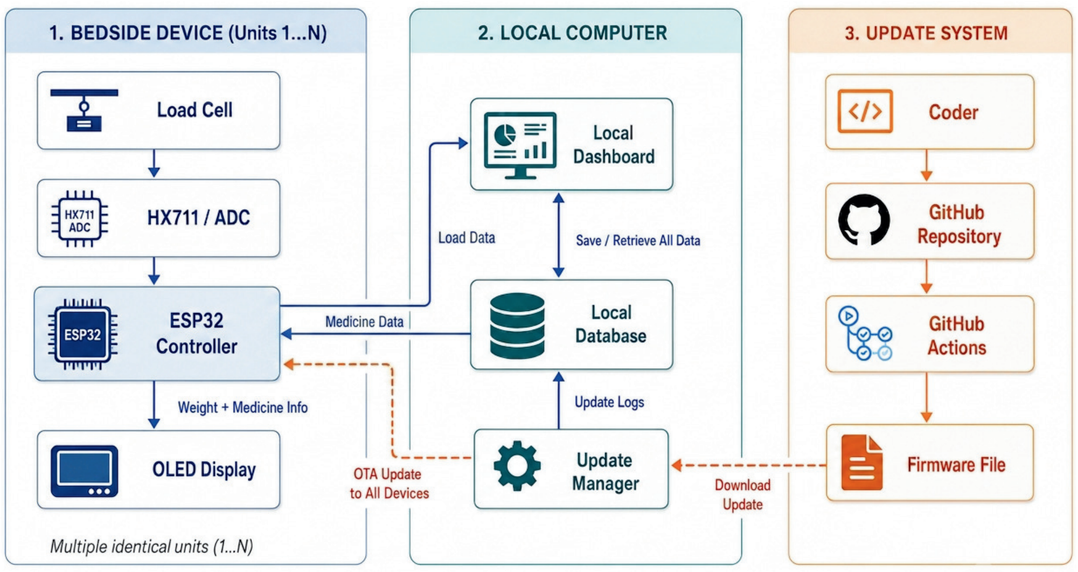
CSE-3200 : Software Development Project - I

Smart IV Fluid Monitoring and Nurse Alert System with Secure OTA Update

Introduction

- This project is about a smart device which automatically monitors the fluid level
- Hospitals often manually check IV fluids leading to missed alerts
- It also alerts staff when is time to give medicine
- A small screen at the bed shows fluid level and medicine info on the device.
- If fluid is low or medicine time comes the screen blinks until a nurse responds
- All beds at a glance, a central screen in the nurse room
- The system can also safely update itself over the internet

Methodology



Expected Outcome

- Nurses get faster & clearer alerts when action is needed
- Fewer missed doses of medicine, empty bags of fluids
- A place to see where all the beds are for nurses
- Reduced time spent on manual bed to bed checks
- Faster response times = safer patients
- System updates itself without a technician on site
- Design low cost for use in many hospitals

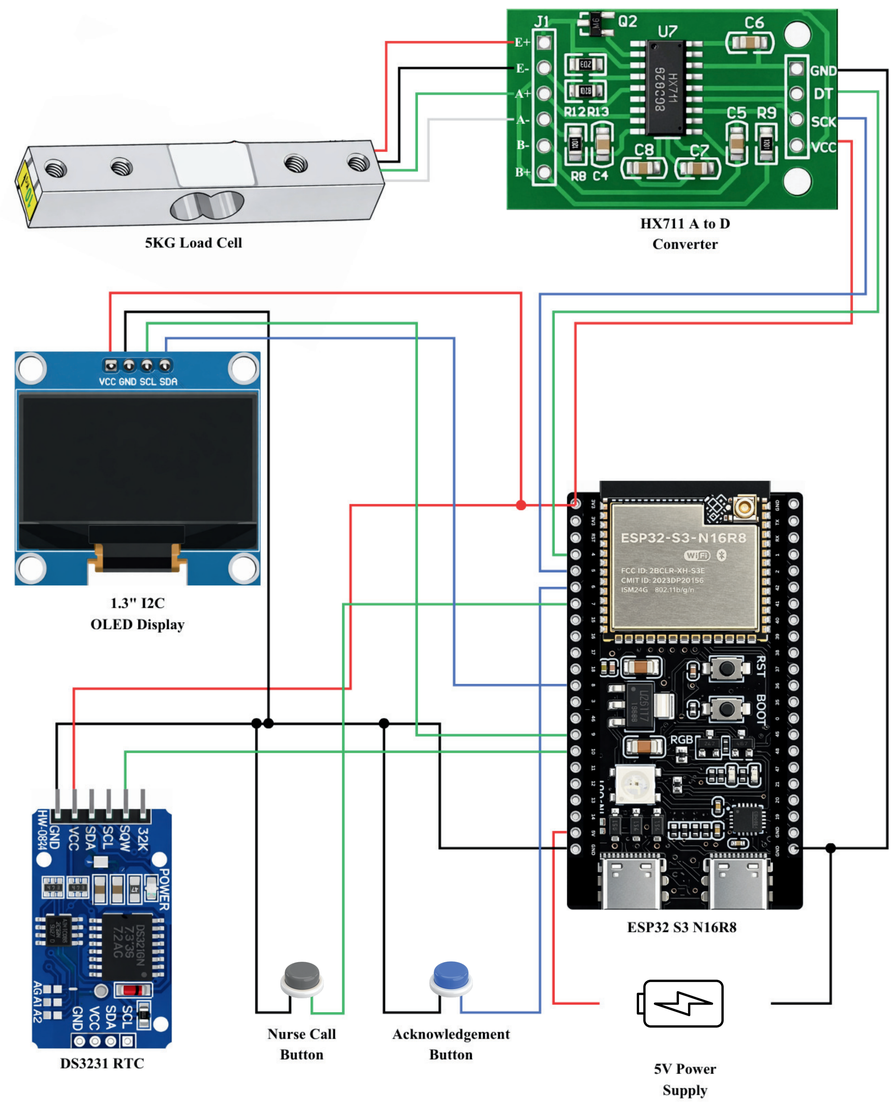
Team Members

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Circuit Diagram



Future Work

- Interface the system with hospital patient data
- Send alerts to nurses' phones, not just the room screen.
- Add backup power in the event of power failure
- Spot problems sooner with smart predictions
- Minimise the number of wires at each bed.
- More health tracking.
- Allow physicians to remotely monitor patients

Conclusion

- Links bedside care with a central monitoring room.
- Solves a real, everyday hospital problem.
- Designed for robust availability, even during upgrades.
- Inexpensive and usable in hospitals with limited resources.
- Eases nurses' workload and improves patient safety.
- Integrates hardware, software and design into one solution.
- A foundation for future smart-hospital systems.